Assignment 2 - Write-up 23L0993

Part A Netflix Movies and TV Shows

Overview. this dataset has 8,807 titles and 12 columns. each row is one Movie or TV Show on Netflix. the columns are type, title, director, cast, country, date added, release year, rating, duration, genres and description. about 70% of the titles are Movies. some columns, like country and genres, can have more than one value in one cell.

What was wrong and what I did

Problem: 3 ratings were really durations (74, 84, 66 min)

Decision: Moved them to duration. Rating left empty.

Justification: The values were right but in the wrong column.

Problem: Director, cast and country were missing (30%, 9%, 9%)

Decision: Filled with 'Unknown'.

Justification: Most missing directors are TV Shows. Dropping rows would lose too much data.

Problem: 10 rows had no date_added

Decision: left them empty.

Justification: I cannot guess a date.

Problem: NR and UR. Spaces in date_added. Extra commas in country.

Decision: Merged NR and UR into 'Unrated'. Removed the spaces and commas.

Justification: NR and UR mean the same. I needed this to fix dates and count countries.

Problem: duration had minutes and seasons together

Decision: split into movie_minutes and seasons.

Justification: I cannot work with numbers stored as text.

Problem: very long movies, many seasons, 14 titles added before release year

Decision: kept them.

Justification: they look like real titles not mistakes.

Insights

1. Movies are most of the catalogue, but TV Shows grew a little.

Observation: Movies are 69.6% of titles. TV Shows were 28.6% of the titles added up to 2018 and 31.4% after 2018. **Interpretation:** TV Shows grew by about 3 points, but Movies are still the majority. **Speculation:** Netflix may have made more series after 2018. The data cannot tell us this.

2. Missing director means something.

Observation: 2,634 titles (29.9%) have no director. 2,446 of them (92.9%) are TV Shows. **Interpretation:** TV Shows often have no single director. So 'Unknown' is about the type of title, not bad data.

3. Most titles are recent and are added fast.

Observation: The median release year is 2017. The median gap between release and adding is 1 year. 36.8% were added in the same year they were released. 24.1% were added more than 5 years after release. Movies have a median gap of 2 years and TV Shows 0. **Interpretation:** Most content is new when it arrives, and TV Shows arrive faster than Movies. gap_years and release_year are almost the same information (r = -0.98). **Speculation:** Some old movies may come from licence deals. The data cannot show this.

4. Kids' movies are shorter.

Observation: The median movie length is 98 minutes. By rating group: Teens 103, Adults 98, Unrated 90, Kids 69.5. Movie length and release year are only weakly related (r = -0.21). **Interpretation:** Rating group helps explain movie length. Release year does not explain much.

5. Different countries have different genres.

Observation: The US has the most titles (3,690), then India (1,046). India's top genres are International Movies (31%) and Dramas (24%). The US is more mixed: Dramas 12%, Comedies 10% and Documentaries 8%. **Interpretation:** Country tells us something about genre. But labels like International Movies already say where a title is from, so the two overlap.

Handoff note for Assignment 3

Useful columns: type, release_year, movie_minutes, seasons, rating_group, num_genres, num_cast, primary_genre and primary_country. Repeated information: release_year, gap_years and release_era (r = -0.98 for the first two), so keep only one. Also rating and rating_group, duration and movie_minutes/seasons, date_added and year_added/month_added, and country and primary_country/num_countries. Do not use as features: show_id, title and description. Director, cast, country and listed_in are long text with many different values. Cleaning notes: I filled director, cast and country with 'Unknown' (30%, 9%, 9%). Unknown director almost means TV Show. movie_minutes is empty for TV Shows and seasons is empty for Movies. 10 titles have no date, so year_added and gap_years are empty. I kept the outliers (movies from 3 to 312 minutes, up to 17 seasons). A log transform could help later. num_genres never goes above 3.

Part B - Supermarket Sales

Overview. This dataset has 1,000 rows and 17 columns. It has sales from 3 branches (Yangon, Mandalay and Naypyitaw) from January to March 2019. Each row is one sale of one product line to one customer. The columns include price, quantity, the money columns, date, time, payment, customer type and a customer rating.

What was wrong and what I did

Problem: No missing values, duplicates or wrong values

Decision: Changed nothing.

Justification: The checks in B2 show everything is fine.

Problem: Date and Time were text

Decision: Changed Date to a date. Took the hour from Time.

Justification: I need these for the time questions.

Problem: Branch and City are the same. The money columns come from each other. Gross margin is the same everywhere.

Decision: Kept all of them. Marked them as repeated.

Justification: The data is correct. I did not want to change the data I was given.

Problem: 9 high Total values (Quantity 10, price about 95 to 99)

Decision: Kept them.

Justification: They are real big purchases, not mistakes.

Problem: I needed time columns

Decision: Made month, day_of_week, is_weekend, hour and time_period.

Justification: I use them for the weekday and time-of-day questions.

Insights

1. The money columns are the same information.

Observation: Cost of goods sold = Unit price x Quantity. Tax 5% is 5% of that. Total is both added. Gross income is the same as Tax 5% in all 1,000 rows. Gross margin is 4.76 in every row. **Interpretation:** One money column is enough. Gross income is just the tax, so it is not real profit.

2. The product line ranking changes with the measure.

Observation: Food and beverages is first for revenue (56,145) but 4th for average sale (322.7). Home and lifestyle is 5th for revenue but 1st for average sale (336.6). Fashion accessories has the most sales (178) but the lowest average (305.1). **Interpretation:** Revenue depends on how many sales there are and how big each sale is. The top five lines are within about 4% of each other.

3. Branch and customer type differences are small.

Observation: Average Total: A 312.3, B 319.9, C 337.1. The spread inside each branch (std) is 232 to 263. Members spend 327.8 and Normal customers 318.1. Members spend more in A and B,

but not in C (336.6 vs 337.7). **Interpretation:** The gaps are small compared to the spread inside each group. The boxplots overlap almost fully. So the differences are small in real life.

4. Sales are highest at 19:00 and on Saturdays.

Observation: 19:00 has the highest total (39,700 from 113 sales). Saturday is highest (56,121 from 164 sales) and Monday is lowest (37,899 from 125 sales). The average sale only goes from 303 to 342 across the weekdays. **Interpretation:** Busy times come from more sales, not bigger sales. Sales per hour are about the same in each time period. Each weekday appears only 12 or 13 times, so some of this may be random.

5. Rating is not related to anything I checked.

Observation: Correlation of rating with Unit price is -0.01, with Quantity -0.02 and with Total -0.04. The average rating is 6.84 to 7.11 across product lines and 6.82 to 7.07 across branches. **Interpretation:** Rating cannot be guessed from these columns. **Speculation:** It may depend on things that are not in the data, like service or waiting time. The data cannot confirm this.

Handoff note for Assignment 3

Useful columns: Unit price and Quantity (they decide the sale value), Product line, Branch, Customer type, Payment, hour and day_of_week. Repeated information: Tax 5%, Cost of goods sold, Gross income and Total are the same information (r = 1.00), so keep only one. Branch and City are also the same. Do not use as features: Invoice ID and Gross margin percentage (same value in every row). The customer rating has almost no link with anything ($|r| < 0.05$), so it is a weak feature. Cleaning notes: I did not remove or change any values. Date became a date and I added 5 columns. I kept the 9 high Total values. Total is skewed (0.89), so a log transform could help.